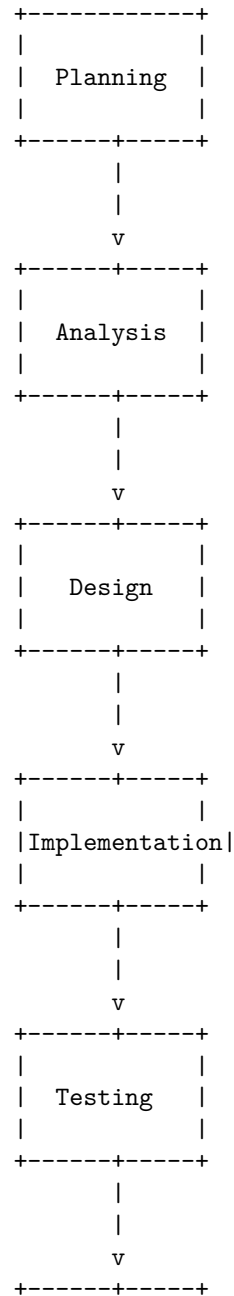


## Software Engineering

Here is an ASCII diagram that represents the software development life cycle, a key concept in software engineering:



```

|           |
| Maintenance |
|           |
+-----+

```

## Unit 1 - Introduction to Software Engineering

```

+-----+
|                                     |
|               Software Engineering |
|                                     |
| +-----+ +-----+ |
| | Analysis | | Design  | |
| +-----+ +-----+ |
|                                     |
| +-----+ +-----+ |
| | Coding   | | Testing  | |
| +-----+ +-----+ |
|                                     |
| +-----+ +-----+ |
| | Debugging| | Maintenance| |
| +-----+ +-----+ |
|                                     |
+-----+

```

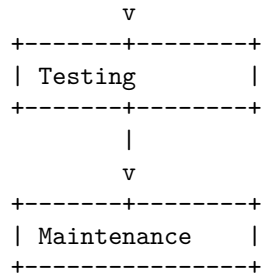
### Introduction to Software Engineering

Software engineering is the process of designing, creating, and maintaining software. It involves the application of engineering principles to the development of software. Here is an ASCII diagram that illustrates the software development process:

```

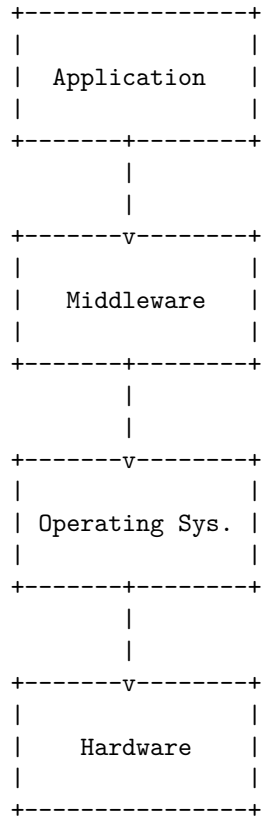
+-----+
| Requirements |
| Analysis     |
+-----+
|
| v
+-----+
| Design       |
+-----+
|
| v
+-----+
| Implementation |
+-----+
|

```



### Software Components

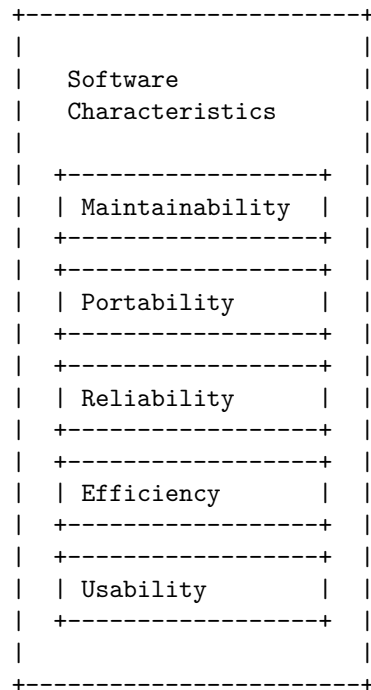
Here is an ASCII diagram that represents the software components of a typical system:



The diagram shows the hierarchy of software components in a typical system. At the top is the application layer, which is the user-facing component of the system. Below that is the middleware layer, which provides services to the application layer and acts as an intermediary between the application and the operating system. The operating system is responsible for managing the hardware resources

of the system and providing services to the application and middleware layers. At the bottom is the hardware layer, which represents the physical components of the system.

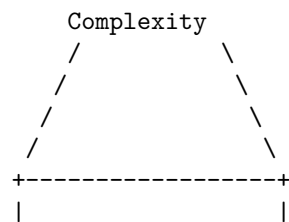
### Software Characteristics

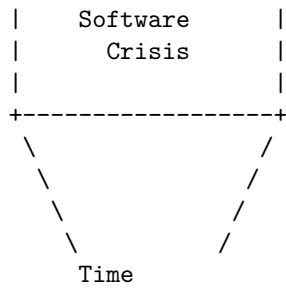


### Software Crisis

The software crisis was a term used in the early days of computing science for the difficulty of writing useful and efficient computer programs in the required time. The software crisis was due to the rapid increases in computer power and the complexity of the problems that could not be tackled. With the increase in the complexity of the software, many software problems arose because existing methods were insufficient.

Here is an ASCII diagram that represents the software crisis:

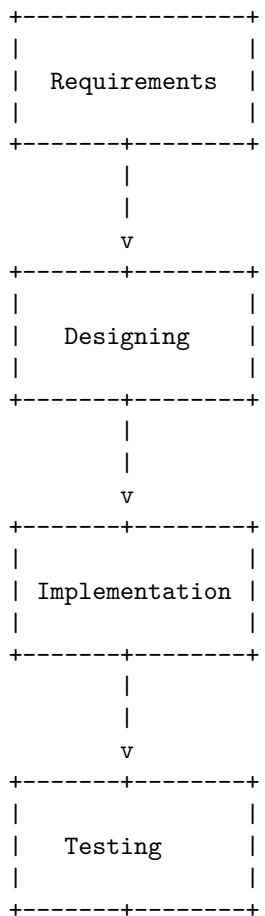


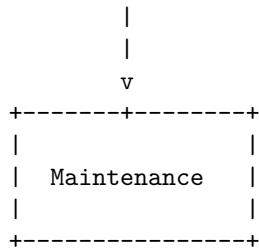


The x-axis represents time, and the y-axis represents the complexity of software. As time progresses, the complexity of software increases, leading to a software crisis.

Sure, here is an ASCII diagram of the Software Engineering Processes:

### Software Engineering Processes





### Similarity and Differences from Conventional Engineering Processes

Here is a detailed ASCII diagram that illustrates the similarities and differences between conventional engineering processes and other processes:

| Conventional Engineering Processes | Other Processes |
|------------------------------------|-----------------|
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |
|                                    |                 |

### Software Quality Attributes

| Software Quality Attributes                                             |                 |  |
|-------------------------------------------------------------------------|-----------------|--|
| <table> <tr> <th>Maintainability</th></tr> <tr> <td></td></tr> </table> | Maintainability |  |
| Maintainability                                                         |                 |  |
|                                                                         |                 |  |
| <table> <tr> <th>Reliability</th></tr> <tr> <td></td></tr> </table>     | Reliability     |  |
| Reliability                                                             |                 |  |
|                                                                         |                 |  |

|             |
|-------------|
|             |
|             |
| Usability   |
|             |
|             |
| Efficiency  |
|             |
|             |
| Portability |
|             |
|             |
| Scalability |
|             |

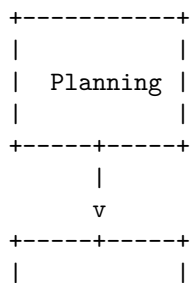
### Software Development Life Cycle (SDLC) Models

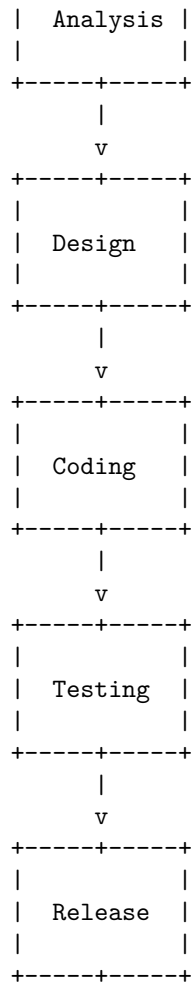
|                      |           |             |         |            |
|----------------------|-----------|-------------|---------|------------|
|                      |           |             |         |            |
| Requirement Analysis | Designing | Development | Testing | Deployment |
|                      |           |             |         |            |

### Water Fall Model in SDLC

The Waterfall Model is a linear sequential approach to software development. It is also known as a classic life cycle model. In this model, the software development process is divided into different phases, and each phase must be completed before moving on to the next one.

Here is an ASCII diagram of the Waterfall Model:

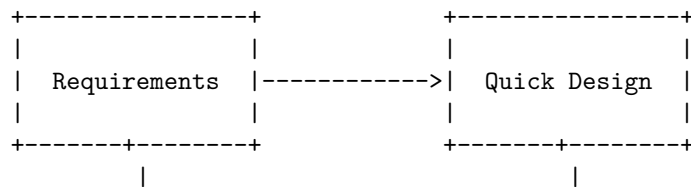


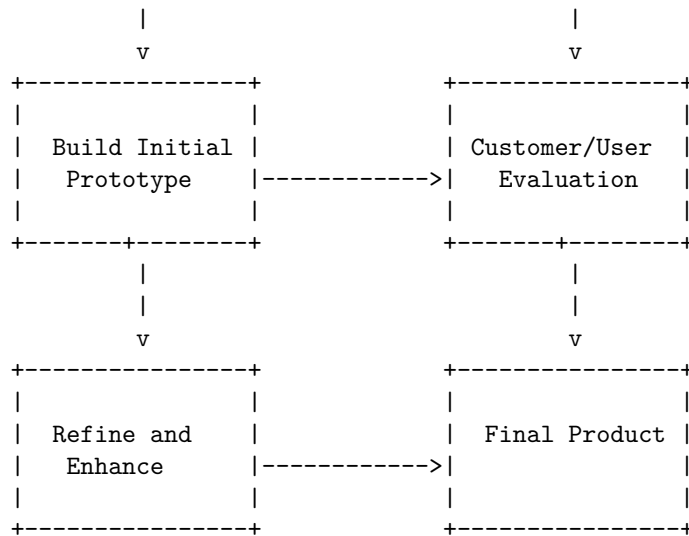


Each phase of the Waterfall Model is distinct and separate. The output of one phase serves as the input for the next phase. The model is called “Waterfall” because the progress flows steadily downwards, like a waterfall, through the phases of Planning, Analysis, Design, Coding, Testing, and Release.

Sure, here is an ASCII diagram for the Prototype Model in SDLC:

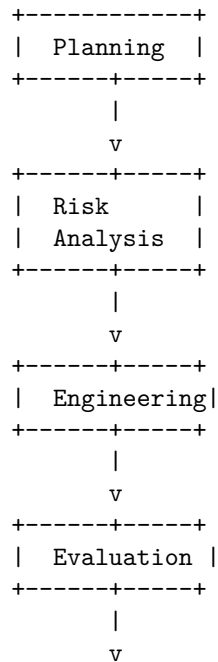
### Prototype Model in SDLC

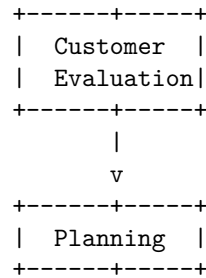




The Spiral Model is a software development process that combines elements of both design and prototyping-in-stages, in an effort to combine advantages of top-down and bottom-up concepts. Here is an ASCII diagram of the Spiral Model in SDLC:

#### Spiral Model in SDLC

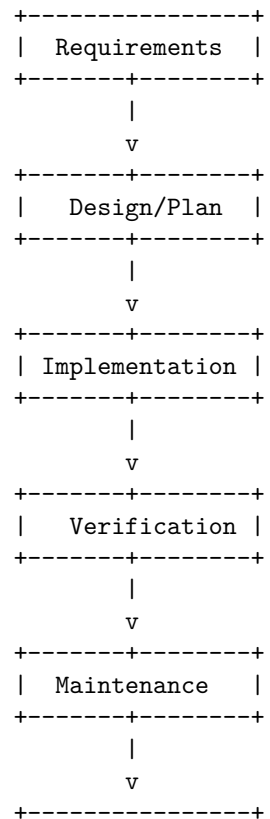




### Evolutionary Development Models in SDLC

Evolutionary development models are iterative and incremental approaches to software development. They involve the development of a system through repeated cycles and in smaller portions at a time, allowing software developers to take advantage of what was learned during the development of earlier parts or versions of the system.

Here is an ASCII diagram that illustrates the process of evolutionary development models in SDLC:

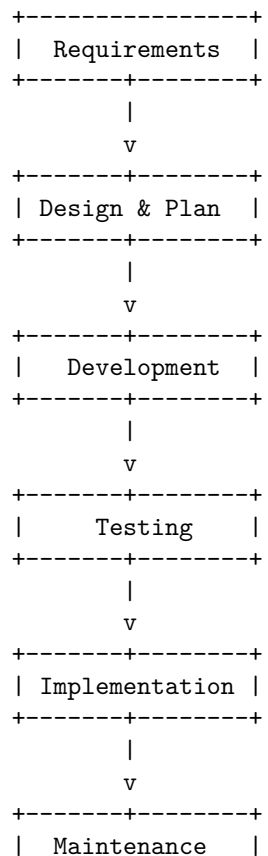


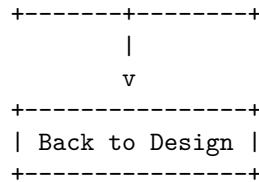
```
| Next Version |
+-----+
```

In this model, the development process starts with gathering and defining requirements. Then, the design and planning phase takes place, followed by the implementation of the software. After the implementation, the software is verified and tested to ensure it meets the requirements. If any issues are found, they are fixed during the maintenance phase. Once the software is stable, the process starts again for the next version of the software, incorporating any new requirements or changes that have been identified.

This model allows for flexibility and adaptability, as changes and new requirements can be incorporated into the development process as they arise. It also allows for the delivery of a working product to the customer earlier in the development process, as each iteration produces a usable version of the software. However, it can also be more difficult to manage and may require more resources than other development models.

### Iterative Enhancement Models in SDLC

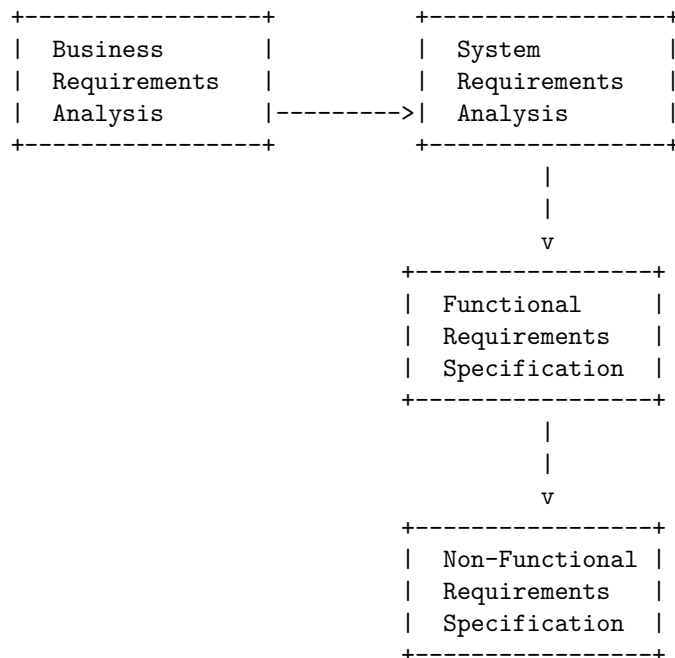




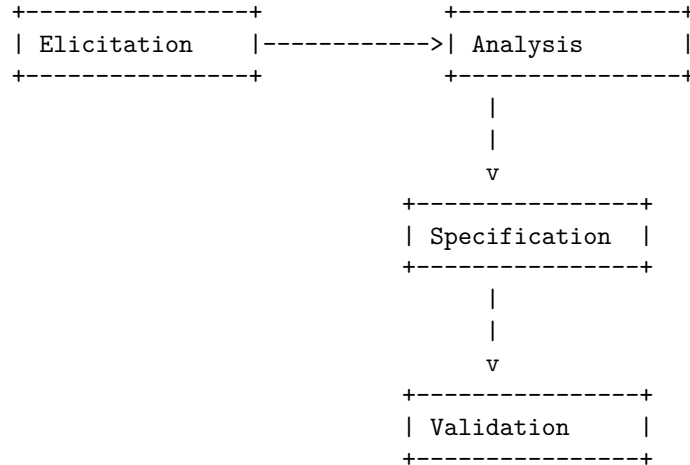
The Iterative Enhancement Model in SDLC is an approach to software development where the software is developed and delivered in small, incremental stages. The process starts with the requirements gathering and analysis phase, followed by the design and planning phase. The development phase is where the actual coding takes place, followed by the testing phase to ensure that the software meets the requirements and is free of defects. The software is then implemented and released to the users. The maintenance phase involves fixing any issues that arise and making any necessary updates. The process then goes back to the design phase to plan for the next iteration of development. This cycle continues until the software is complete and meets all the requirements. This model allows for flexibility and the ability to make changes and improvements throughout the development process.

## Unit 2 - Software Requirement Specifications (SRS)

Here is an ASCII diagram for the Software Requirement Specifications (SRS) process:

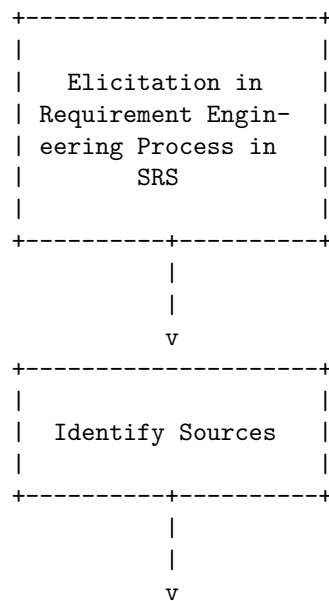


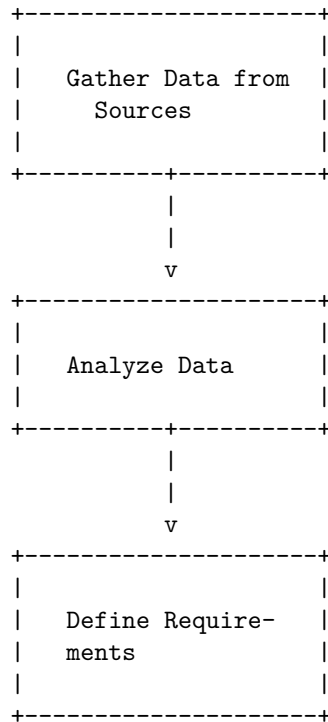
## Requirement Engineering Process in SRS



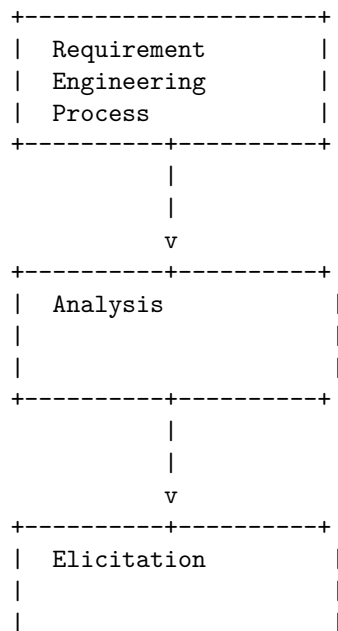
The Requirement Engineering Process in SRS consists of four main steps: Elicitation, Analysis, Specification, and Validation. During the Elicitation phase, requirements are gathered from stakeholders. In the Analysis phase, the requirements are analyzed for consistency and completeness. In the Specification phase, the requirements are documented in a Software Requirements Specification (SRS) document. Finally, in the Validation phase, the requirements are validated to ensure that they meet the needs of the stakeholders.

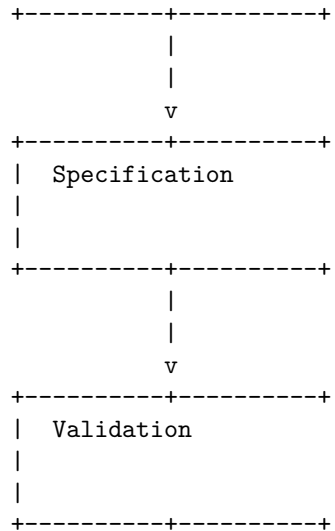
## Elicitation in Requirement Engineering Process in SRS



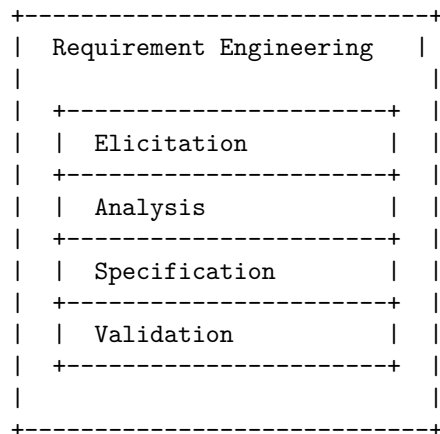


### Analysis in Requirement Engineering Process in SRS

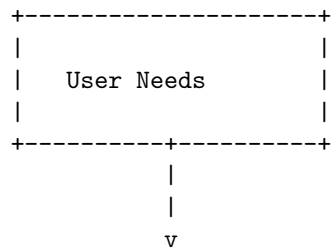


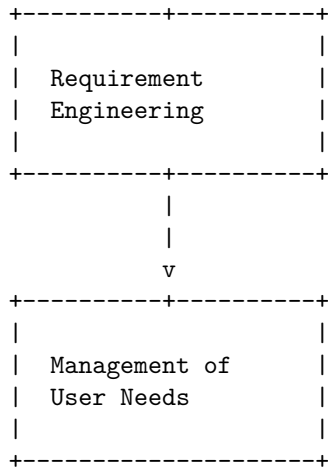


#### Documentation in Requirement Engineering Process in SRS



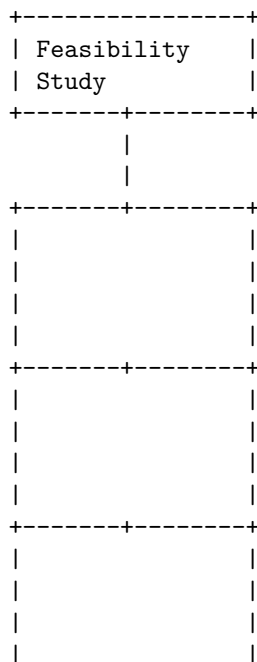
#### Review and Management of User Needs in Requirement Engineering Process in SRS

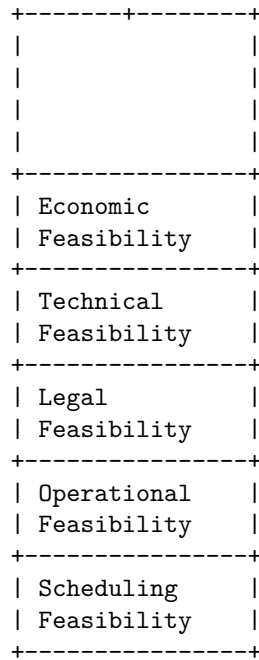




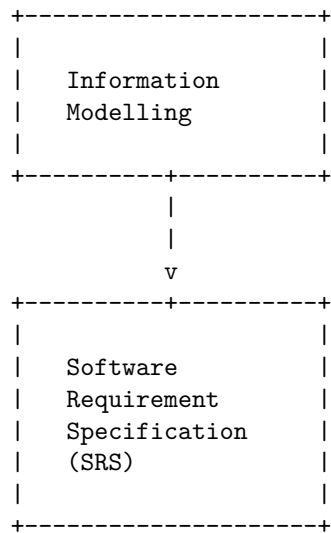
### Feasibility Study in Software Requirement Specification (SRS)

A feasibility study is an analysis that takes all of a project's relevant factors into account—including economic, technical, legal, and scheduling considerations—to ascertain the likelihood of completing the project successfully. Here is an ASCII diagram that represents the different types of feasibility studies that are typically performed as part of a Software Requirement Specification (SRS) document:





Information Modelling in Software Requirement Specification (SRS)

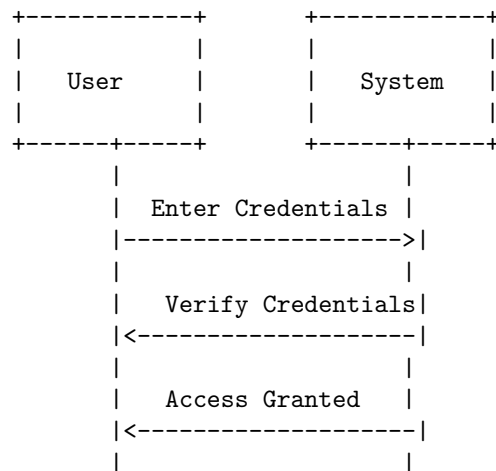


This diagram shows the relationship between Information Modelling and Software Requirement Specification (SRS). Information Modelling is a process of representing the information requirements of a system in a structured way. It is used to define the data structures, relationships, and constraints of the system.

The output of the Information Modelling process is used as an input to the Software Requirement Specification (SRS) process, which defines the functional and non-functional requirements of the system.

### Data Flow Diagrams in Software Requirement Specification (SRS)

A data flow diagram (DFD) is a graphical representation of the flow of data in an information system. It shows how data is input, processed, stored, and output from the system. Here is an example of a DFD for a simple login process:

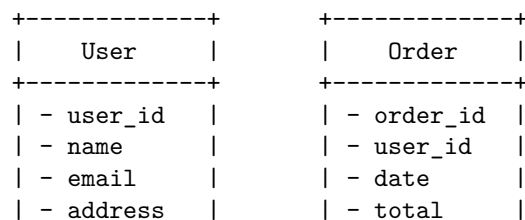


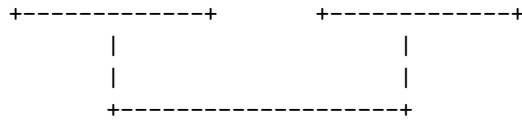
In this diagram, the user enters their credentials, which are then sent to the system for verification. If the credentials are valid, the system grants access to the user. This is a simple example of how a DFD can be used to represent the flow of data in a system.

### Entity Relationship Diagrams in Software Requirement Specification (SRS)

An Entity Relationship Diagram (ERD) is a visual representation of the relationships between entities in a database. It is commonly used in the design of database systems and is an important part of a Software Requirement Specification (SRS) document.

Here is an example of an ERD in ASCII art:





In this example, the **User** entity has attributes such as **user\_id**, **name**, **email**, and **address**. The **Order** entity has attributes such as **order\_id**, **user\_id**, **date**, and **total**. The relationship between the two entities is represented by the line connecting them, indicating that a **User** can have many **Orders**.

### Decision Tables in Software Requirement Specification (SRS)

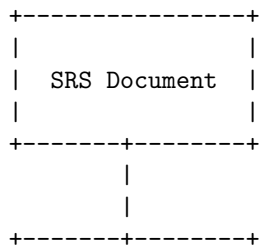
A decision table is a tabular representation of the conditions and actions of a system. It is used to specify the behavior of a system in a structured and organized manner. Here is an example of a decision table:

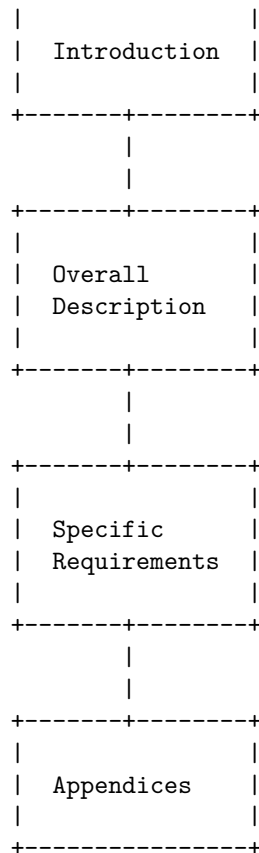
| Condition | Case 1  | Case 2  | Case 3  |
|-----------|---------|---------|---------|
| Input A   | True    | False   | True    |
| Input B   | False   | True    | True    |
| Action X  | Perform | Don't   | Perform |
| Action Y  | Don't   | Perform | Perform |

In this example, the decision table specifies the behavior of a system with two inputs (A and B) and two actions (X and Y). The table shows that when input A is true and input B is false (Case 1), the system should perform action X but not action Y. Similarly, when input A is false and input B is true (Case 2), the system should perform action Y but not action X. When both inputs are true (Case 3), the system should perform both actions.

Decision tables are commonly used in software requirement specification (SRS) documents to specify the behavior of a system in a clear and concise manner. They can help to reduce ambiguity and improve the communication between the development team and the stakeholders.

### SRS Document

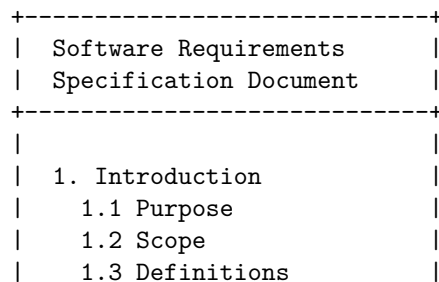




The IEEE Standards for Software Requirements Specification (SRS) is a set of guidelines for creating a document that outlines the requirements of a software system. The standard, IEEE 830-1998, provides a structure for the SRS document and outlines the information that should be included.

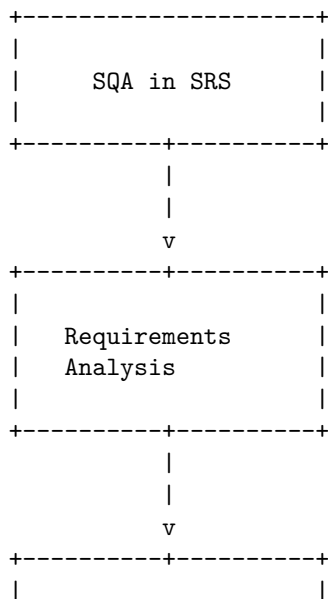
Here is an ASCII diagram that shows the structure of an SRS document according to the IEEE standard:

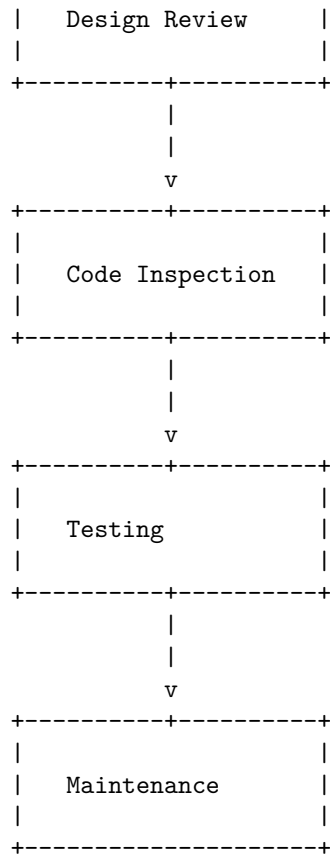
#### IEEE Standards for SRS



|         |                          |  |
|---------|--------------------------|--|
|         | 1.4 References           |  |
|         | 1.5 Overview             |  |
|         | 2. Overall Description   |  |
|         | 2.1 Product Perspective  |  |
|         | 2.2 Product Functions    |  |
|         | 2.3 User Characteristics |  |
|         | 2.4 Constraints          |  |
|         | 2.5 Assumptions          |  |
|         | 3. Specific Requirements |  |
|         | 3.1 Functional           |  |
|         | 3.2 Performance          |  |
|         | 3.3 Interface            |  |
|         | 3.4 Operational          |  |
|         | 3.5 Resource             |  |
|         | 3.6 Verification         |  |
|         | 4. Appendices            |  |
|         | 4.1 Index                |  |
|         | 4.2 Supporting           |  |
|         | 4.3 Information          |  |
|         |                          |  |
| +-----+ |                          |  |

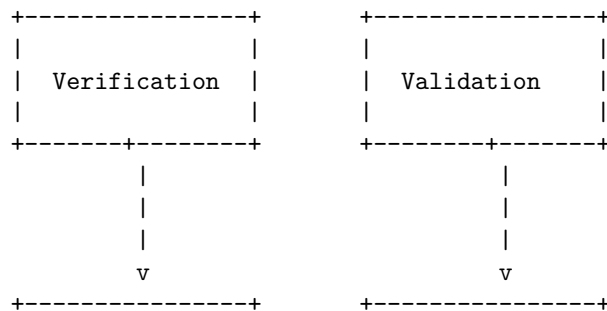
## Software Quality Assurance (SQA) in SRS

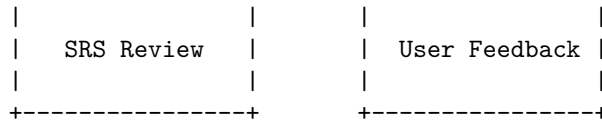




### Verification and Validation in SRS

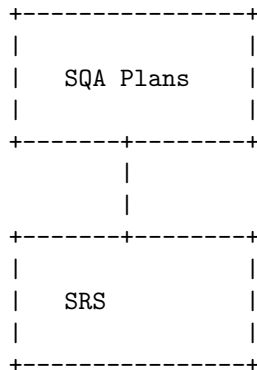
Verification and validation are two important processes in software development. Verification is the process of ensuring that the software meets the specified requirements, while validation is the process of ensuring that the software meets the needs of the user. Here is an ASCII diagram that illustrates the relationship between these two processes:





### SQA Plans in SRS

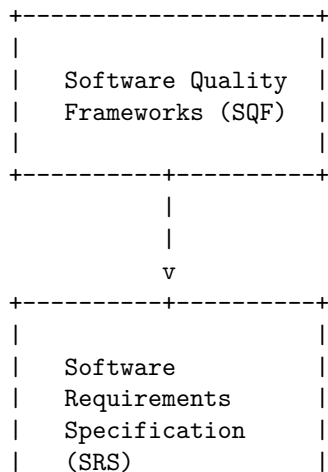
Here is an ASCII diagram for SQA Plans in SRS:



The diagram shows that the SQA Plans are a part of the SRS. The SQA Plans are used to ensure the quality of the software being developed. The SRS, or Software Requirements Specification, is a document that describes the requirements for the software being developed. The SQA Plans are an important part of the SRS, as they help to ensure that the software meets the requirements specified in the SRS.

### Software Quality Frameworks (SQF) in SRS

Here is an ASCII diagram for Software Quality Frameworks (SQF) in SRS:



```

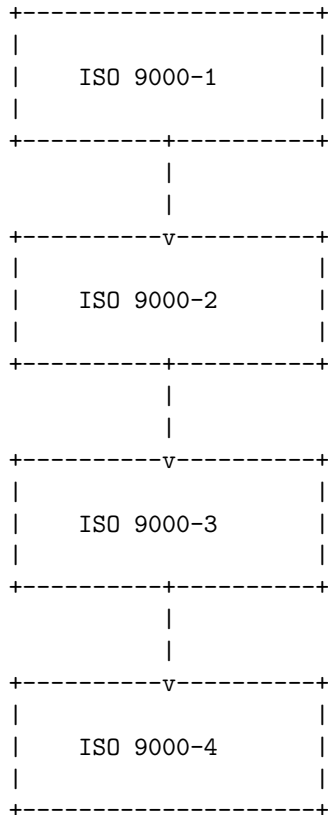
|                                     |
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```

ISO 9000 is a series of standards that define, establish, and maintain an effective quality assurance system for manufacturing and service industries. The ISO 9000 standard is based on the idea of a process approach to quality management. The standard is divided into several parts, including ISO 9001, which specifies the requirements for a quality management system.

Here is an ASCII diagram that shows the relationship between the different ISO 9000 models in the context of a Software Requirements Specification (SRS):

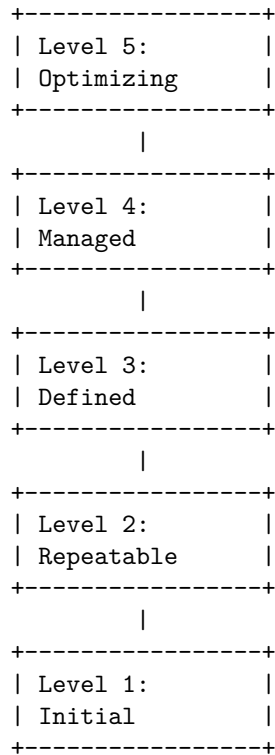
### ISO 9000 Models in SRS



### SEI-CMM Model in SRS

The SEI-CMM (Software Engineering Institute - Capability Maturity Model) is a model used to assess the maturity of software development processes. It consists of five levels, each representing a different stage of maturity.

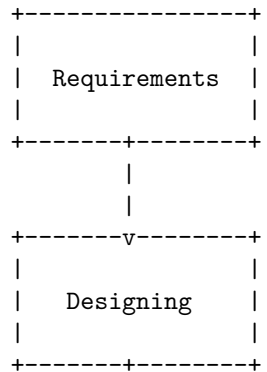
Here is an ASCII diagram representing the SEI-CMM model:

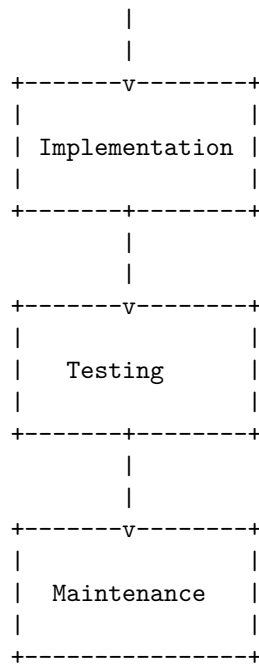


At Level 1, the Initial level, processes are ad hoc and unstructured. At Level 2, the Repeatable level, basic project management processes are established. At Level 3, the Defined level, processes are standardized and documented. At Level 4, the Managed level, processes are measured and controlled. At Level 5, the Optimizing level, processes are continuously improved.

### Unit 3 - Software Design

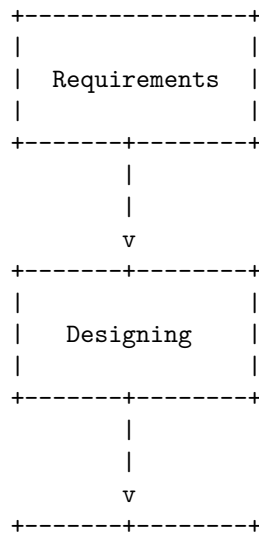
Here is an ASCII diagram for software design:

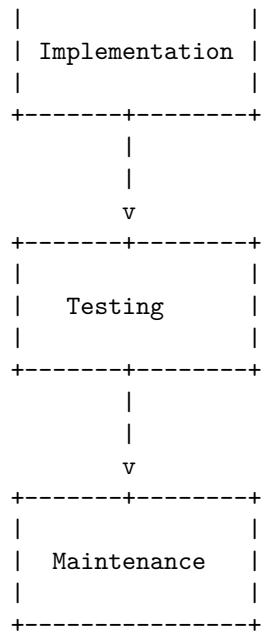




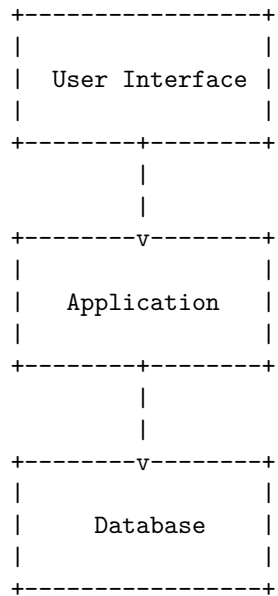
### Basic Concept of Software Design

Software design is the process of defining software methods, functions, objects, and the overall structure and interaction of the code so that the resulting functionality will satisfy the user's requirements. Here is an ASCII diagram that illustrates the basic concept of software design:



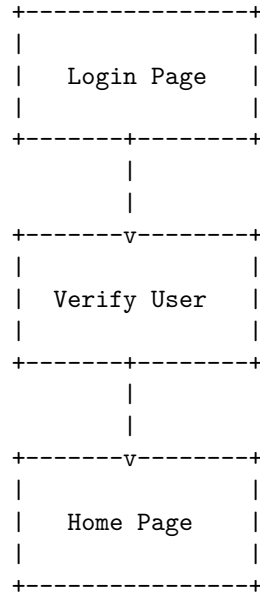


#### Architectural Design in Software Design



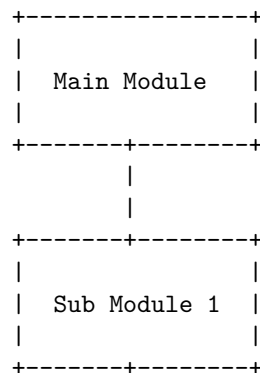
## Low Level Design in Software Design

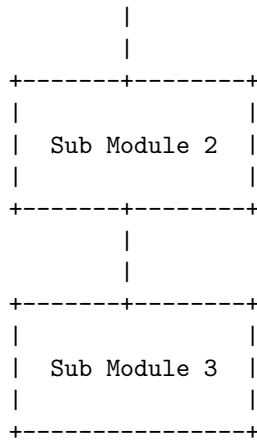
Low-level design (LLD) is a component-level design process that follows a step-by-step refinement process. This process can be graphically represented through a flowchart or pseudocode. Here is an example of an ASCII diagram that represents a low-level design for a login process:



This diagram shows the flow of the login process, starting from the login page, where the user enters their credentials. The next step is to verify the user, which is done by checking the entered credentials against the stored user information. If the verification is successful, the user is directed to the home page. Otherwise, an error message is displayed, and the user is prompted to try again.

## Modularization in Software Design

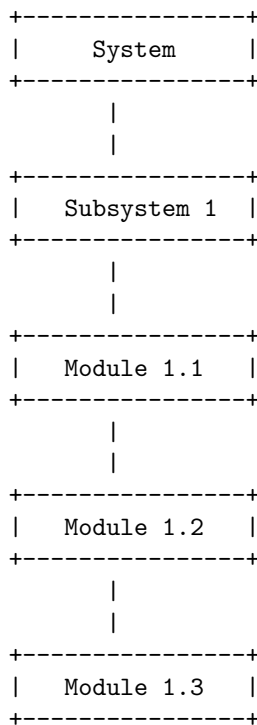




### Design Structure Charts in Software Design

Design Structure Charts (DSCs) are a graphical representation of the design of a software system. They show the hierarchical structure of the system, the modules and their relationships, and the flow of data and control between them.

Here is an example of a Design Structure Chart:



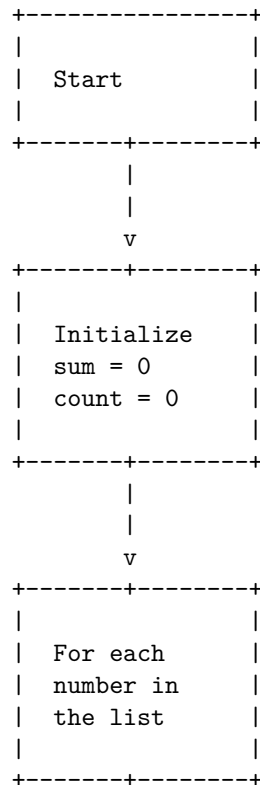
In this example, the system is composed of one subsystem, which is further divided into three modules. The arrows show the flow of data and control between the different levels of the hierarchy.

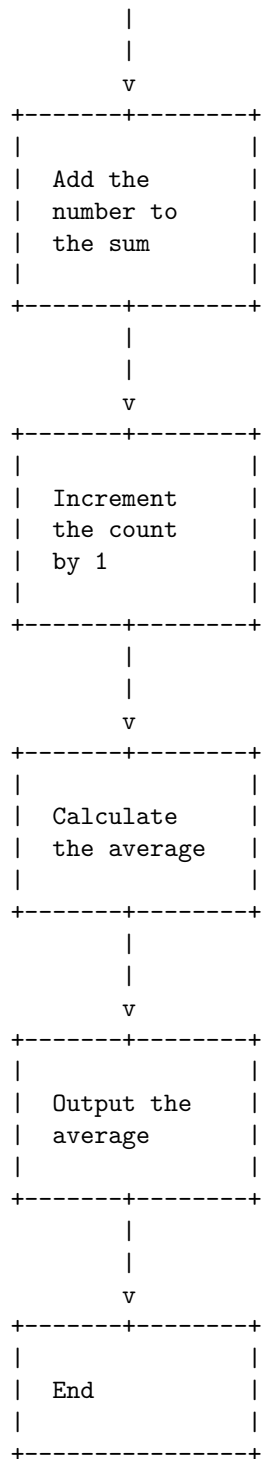
### Pseudo Codes in Software Design

Pseudo code is a way to describe how a computer program or algorithm should work, using natural language instead of a specific programming language. It is often used in software design to help developers visualize the flow of a program before writing actual code. Here is an example of a pseudo code for a simple program that calculates the average of a list of numbers:

1. Initialize a variable called ``sum`` to 0
2. Initialize a variable called ``count`` to 0
3. For each number in the list:
  - a. Add the number to the ``sum``
  - b. Increment the ``count`` by 1
4. Calculate the average by dividing the ``sum`` by the ``count``
5. Output the average

Here is an ASCII diagram that represents the flow of the pseudo code above:

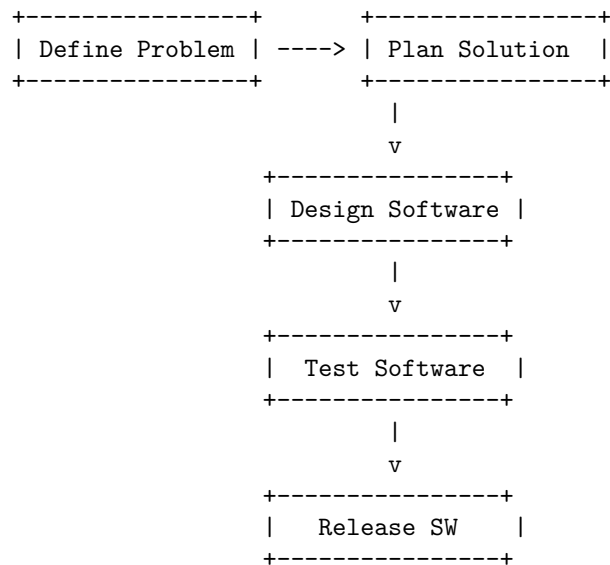




## Flow Charts in Software Design

A flowchart is a visual representation of the sequence of steps and decisions needed to perform a process. Each step in the sequence is noted within a diagram shape. Steps are linked by connecting lines and directional arrows. This allows anyone to view the flowchart and logically follow the process from beginning to end.

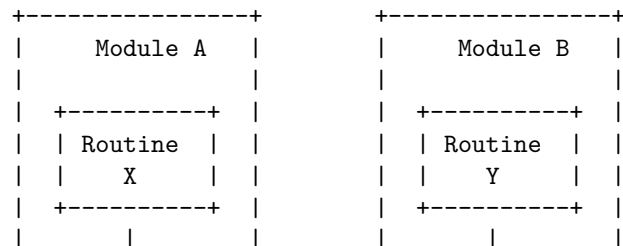
Here is an example of a simple flowchart for the process of designing software:

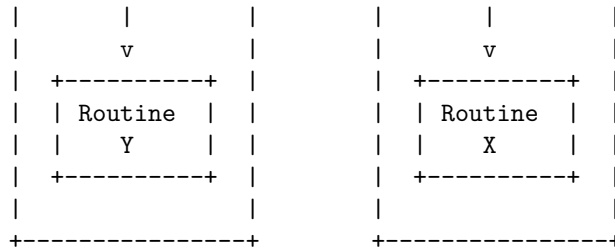


## Coupling in Software Design

Coupling refers to the degree of interdependence between software modules. It is a measure of how closely connected two routines or modules are and the strength of the relationships between them. Low coupling is often a sign of a well-structured computer system and a good design, while high coupling is associated with complex and difficult-to-maintain systems.

Here is an ASCII diagram that illustrates the concept of coupling in software design:





In this diagram, Module A and Module B are two separate modules in a software system. Routine X in Module A calls Routine Y in Module B, and Routine Y in Module B calls Routine X in Module A. This represents a high degree of coupling between the two modules, as changes in one module may affect the other.

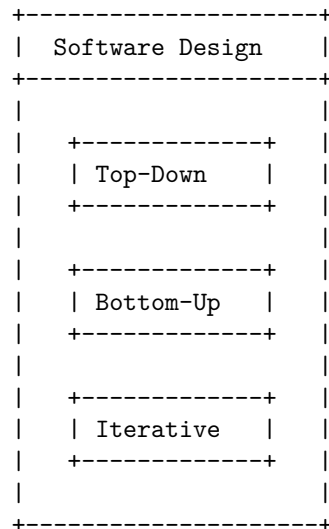
### Cohesion Measures in Software Design

Cohesion refers to the degree to which the elements of a module belong together. In software design, it is considered desirable to have high cohesion. There are several measures of cohesion, including:

| Measure         | Description                                                                      |
|-----------------|----------------------------------------------------------------------------------|
| Coincidental    | Elements grouped together with no meaningful relationship.                       |
| Logical         | Elements grouped together because they are logically related.                    |
| Temporal        | Elements grouped together because they are related in time.                      |
| Procedural      | Elements grouped together because they are part of a procedure.                  |
| Communicational | Elements grouped together because they operate on the same data.                 |
| Sequential      | Elements grouped together because the output of one is the input of another.     |
| Functional      | Elements grouped together because they contribute to a single well-defined task. |

## Design Strategies in Software Design

There are several design strategies in software design, including top-down design, bottom-up design, and iterative design. Here is an ASCII diagram that illustrates these strategies:

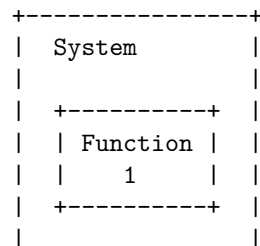


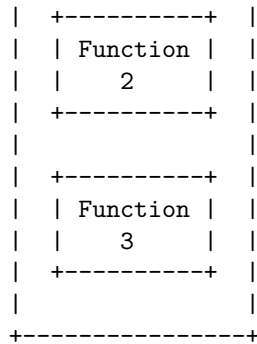
In top-down design, the overall system is first defined, and then broken down into smaller components. In bottom-up design, the individual components are first designed and then combined to form the overall system. In iterative design, the system is designed, tested, and refined in a cyclical process until the desired result is achieved.

## Function Oriented Design in Software Design

Function Oriented Design is a software design methodology that focuses on the decomposition of the system into a set of interacting functions. The main idea behind this approach is to identify the major functions that the system must perform and design the software structure around these functions.

Here is an ASCII diagram that illustrates the basic concept of Function Oriented Design:

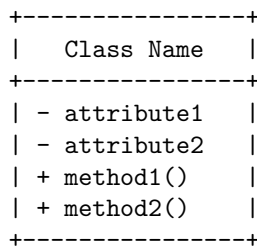




In this diagram, the system is represented as a box that contains several smaller boxes, each representing a function. The functions interact with each other to perform the overall task of the system. The design process involves identifying the functions, determining their interactions, and organizing them in a logical and efficient manner.

### Object Oriented Design in Software Design

Here is an ASCII diagram that represents the basic principles of Object Oriented Design in Software Design:



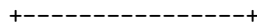
In Object Oriented Design, a class represents an object or a concept in the software. The class has attributes (represented by the - symbol) that describe the characteristics of the object. The class also has methods (represented by the + symbol) that define the behavior of the object.

The diagram above shows a simple class with two attributes and two methods. The class name is at the top of the diagram, and the attributes and methods are listed below it.

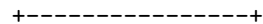
### Top-Down and Bottom-Up Design in Software Design

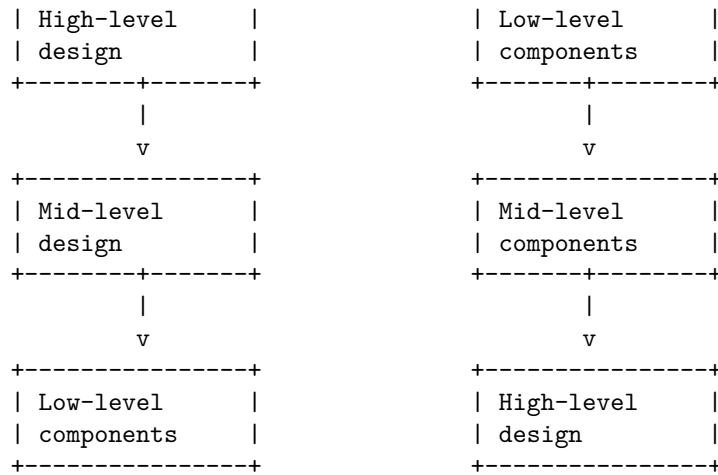
Top-down and bottom-up are two approaches to the design of software. Here is an ASCII diagram that illustrates the difference between the two approaches:

Top-Down:



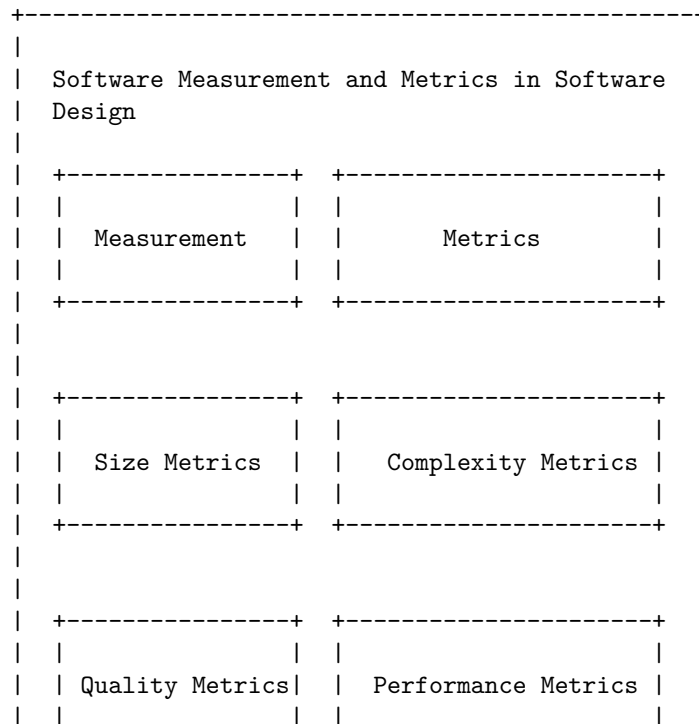
Bottom-Up:





In top-down design, the high-level design is created first, and then broken down into more detailed, lower-level components. In bottom-up design, the low-level components are created first, and then combined to form higher-level components and eventually the overall design.

### Software Measurement and Metrics in Software Design





```

| N1 = total operators |
| N2 = total operands  |
|                       |
+-----+
|       |
| Vocabulary = n1 + n2 |
| Length = N1 + N2     |
| Volume = Length *    |
|         log2(Vocab)  |
|       |
+-----+
|       |
| Difficulty = (n1/2) * |
|               (N2/n2) |
| Effort = Difficulty * |
|               Volume  |
|       |
+-----+

```

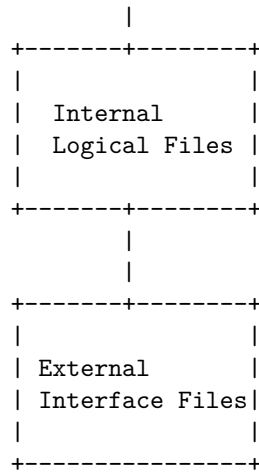
Function Point (FP) Based Measures in software design is a technique used to measure the size and complexity of a software system. It is based on the user's view of the system and takes into account the functionality provided by the system.

Here is an ASCII diagram that illustrates the components of Function Point (FP) Based Measures in software design:

```

+-----+
|       |
| User Inputs |
|       |
+-----+
|       |
|       |
+-----+
|       |
| User Inquiries |
|       |
+-----+
|       |
|       |
+-----+
|       |
| User Outputs |
|       |
+-----+
|       |

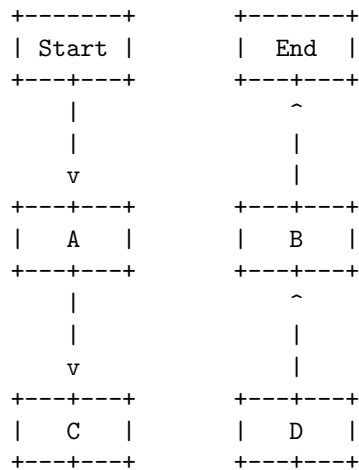
```



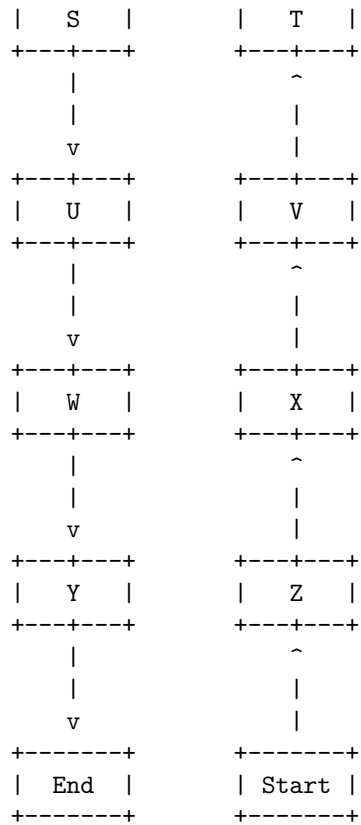
Each of these components represents a different type of functionality provided by the system. User Inputs represent the data that is entered into the system by the user. User Inquiries represent the requests for information made by the user. User Outputs represent the data that is generated by the system in response to user requests. Internal Logical Files represent the data that is stored and maintained within the system. External Interface Files represent the data that is exchanged between the system and other external systems.

Cyclomatic complexity is a software metric used to measure the complexity of a program. It is calculated by developing a Control Flow Graph of the code that measures the number of linearly-independent paths through a program module. Here is an ASCII diagram that represents the Control Flow Graph of a sample code:

### Cyclomatic Complexity Measures in software design

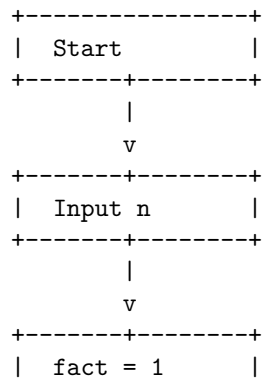


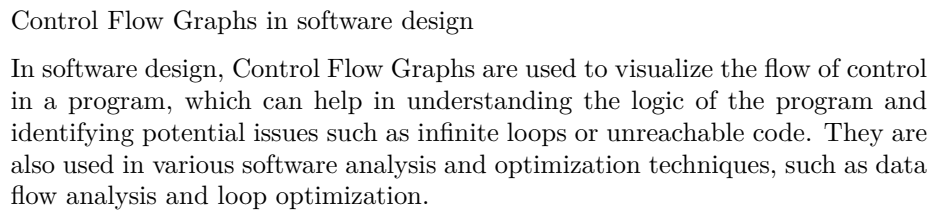
|           |           |
|-----------|-----------|
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| E         | F         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| G         | H         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| I         | J         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| K         | L         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| M         | N         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| O         | P         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |
| Q         | R         |
| +---+---+ | +---+---+ |
|           | ^         |
|           |           |
| v         |           |
| +---+---+ | +---+---+ |



A Control Flow Graph (CFG) is a graphical representation of the control flow of a program. Each node in the graph represents a basic block, which is a sequence of instructions with a single entry point and a single exit point. The edges between the nodes represent the flow of control between the basic blocks.

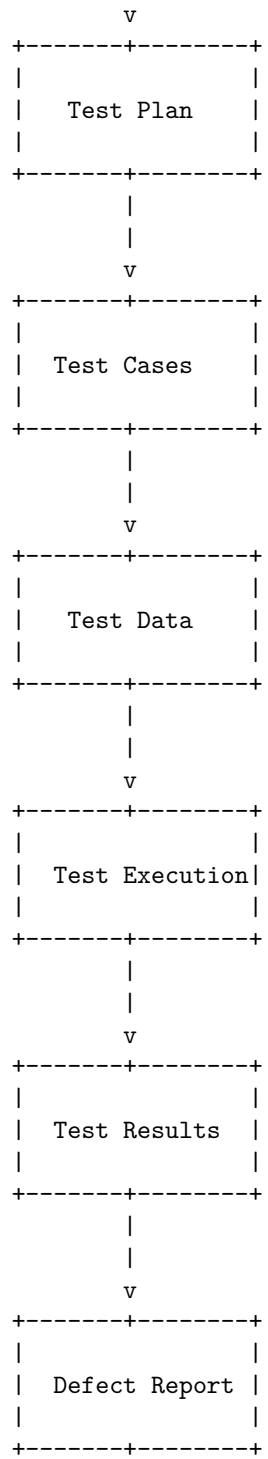
Here is an example of a Control Flow Graph for a simple program that calculates the factorial of a number:

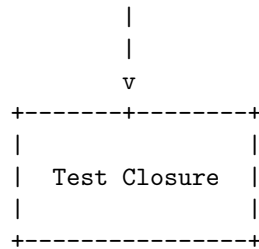




Here is an ASCII diagram that represents the software testing process:

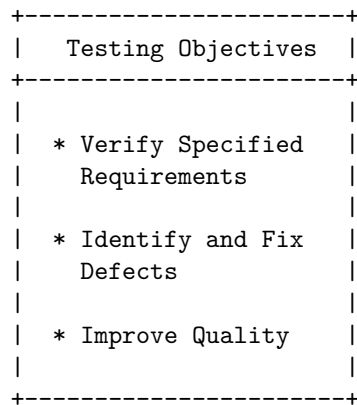






### Testing Objectives in Software Testing

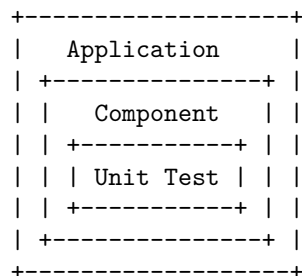
The main objectives of software testing are to ensure that the software meets the specified requirements, to identify and fix defects, and to improve the overall quality of the software. Here is an ASCII diagram that illustrates the testing objectives in software testing:



### Unit Testing in Software Testing

Unit testing is a software testing technique in which individual units or components of a software application are tested in isolation from the rest of the application. The purpose of unit testing is to validate that each unit or component of the software application is working as intended.

Here is an ASCII diagram that illustrates the concept of unit testing:

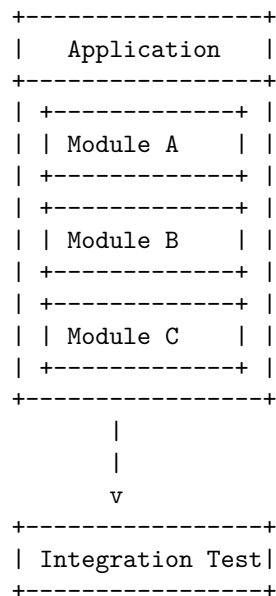


In this diagram, the **Application** represents the entire software application, which is composed of multiple **Components**. Each **Component** has its own set of **Unit Tests** that are designed to test the functionality of that specific component in isolation from the rest of the application.

Unit testing is typically performed by developers as part of the development process, and it is considered a best practice in software development as it helps to catch issues early on in the development process, before the application is released to users. Unit tests are typically automated, and they can be run every time the code is changed to ensure that the changes did not introduce any new issues.

### Integration Testing in Software Testing

Integration testing is a level of software testing where individual units are combined and tested as a group. The purpose of this level of testing is to expose faults in the interaction between integrated units. Here is an ASCII diagram that illustrates the concept of integration testing:



### Acceptance Testing in Software Testing

Acceptance testing is a level of software testing where a system is tested for acceptability. The purpose of this test is to evaluate the system's compliance with the business requirements and assess whether it is acceptable for delivery.

Here are some key points to remember about acceptance testing:

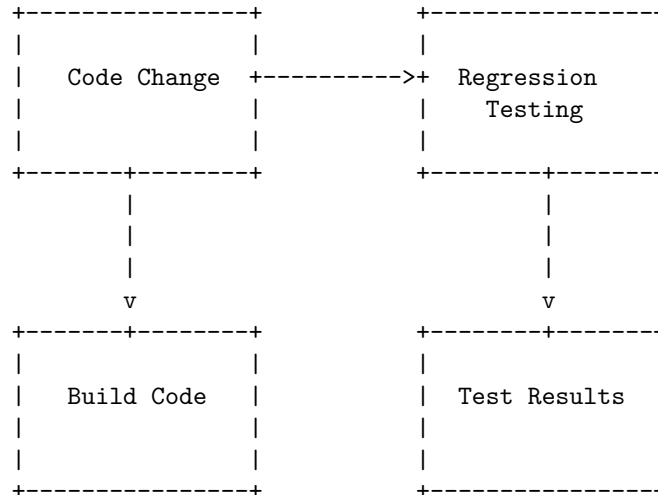
1. Acceptance testing is performed by the customer, the user, or a subject

matter expert.

2. It is the final stage of testing before the software is released to the market.
3. Acceptance testing can be either functional or non-functional.
4. The goal of acceptance testing is to ensure that the software meets the customer's needs and expectations.
5. Acceptance testing can be performed either on-site or off-site.
6. Acceptance testing can be either formal or informal.
7. Acceptance testing is also known as user acceptance testing (UAT), end-user testing, or field testing.

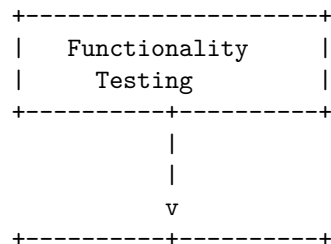
### Regression Testing in Software Testing

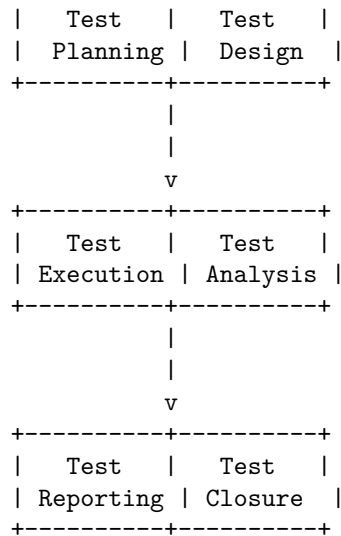
Here is an ASCII diagram that illustrates the process of regression testing in software testing:



In this diagram, a code change triggers the process of regression testing. The changed code is built and then tested to ensure that the changes did not introduce any new bugs or regressions. The test results are then analyzed to determine if any further action is needed.

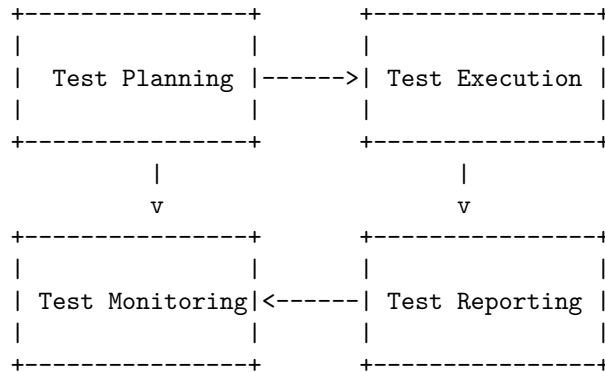
### Testing for Functionality in Software Testing





### Testing for Performance in Software Testing

Performance testing is a type of software testing that is used to determine the speed, responsiveness, and stability of a system under a particular workload. Here is an ASCII diagram that illustrates the process of performance testing:



The first step in performance testing is test planning, where the objectives and goals of the test are defined. Next, the test is executed and monitored to collect data on the system's performance. Finally, the results of the test are reported and analyzed to determine if the system meets the performance requirements.

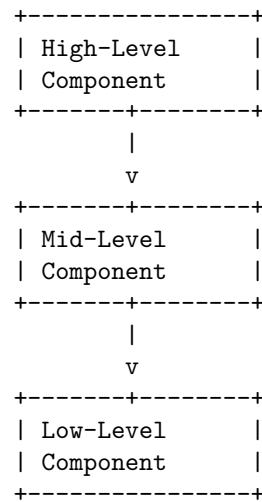
### Top-Down and Bottom-Up Testing Strategies in Software Testing

Top-down and bottom-up are two approaches to testing software. Top-down testing starts with the highest level of the software architecture and works its way down, while bottom-up testing starts with the lowest level and works its

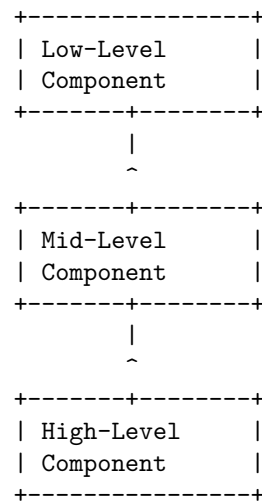
way up.

Here is an ASCII diagram that illustrates the difference between the two approaches:

Top-Down Testing:

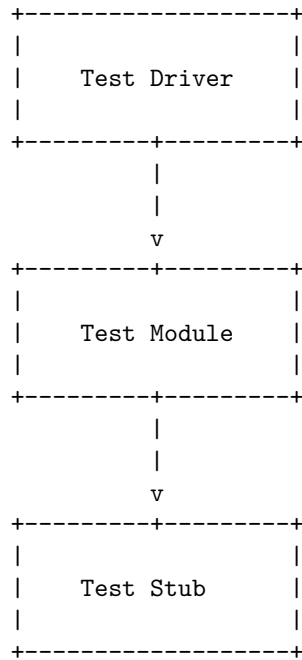


Bottom-Up Testing:

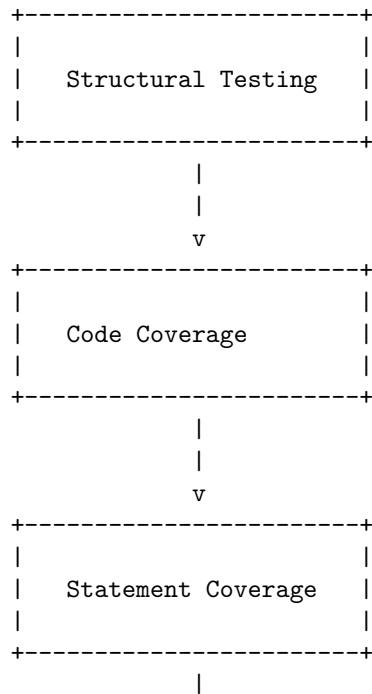


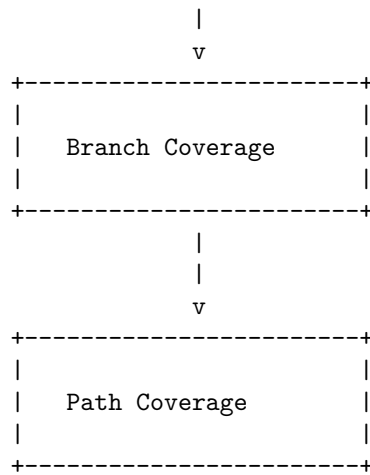
In top-down testing, the high-level component is tested first, followed by the mid-level component, and finally the low-level component. In bottom-up testing, the low-level component is tested first, followed by the mid-level component, and finally the high-level component.

### Test Drivers and Test Stubs software testing strategy

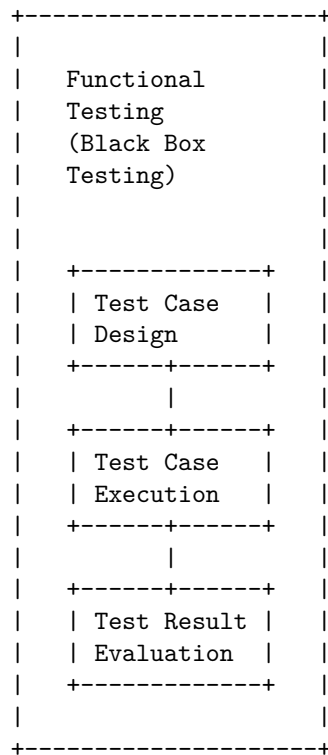


#### Structural Testing (White Box Testing) software testing strategy





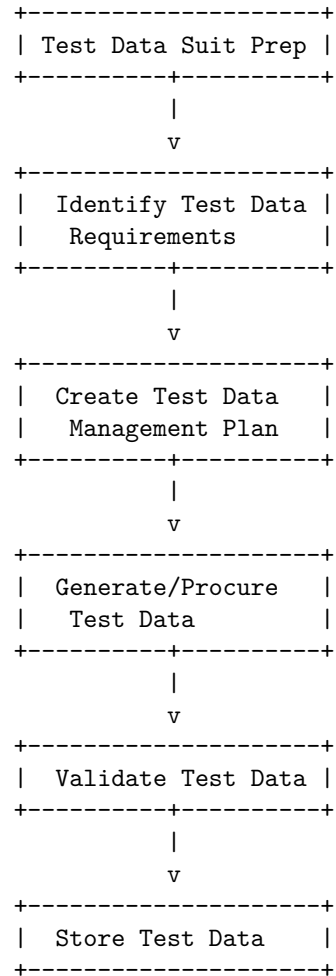
#### Functional Testing (Black Box Testing) software testing strategy



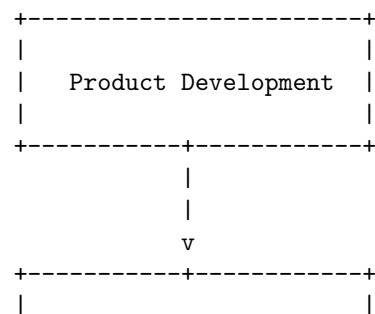
#### Test Data Suit Preparation software testing strategy

Here is an ASCII diagram that represents the Test Data Suit Preparation soft-

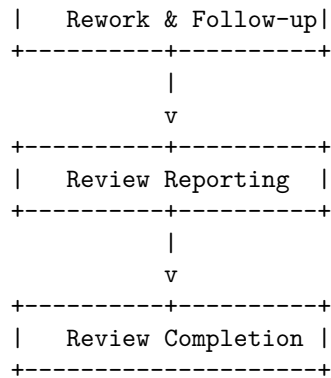
ware testing strategy:



Alpha and Beta Testing of Products software testing strategy



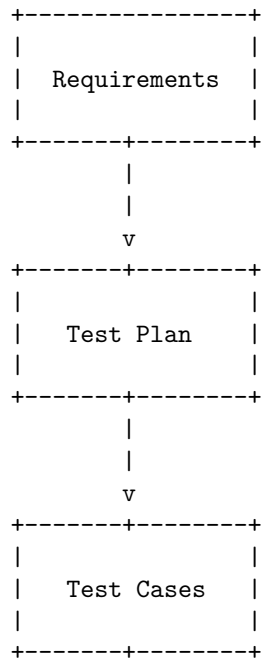


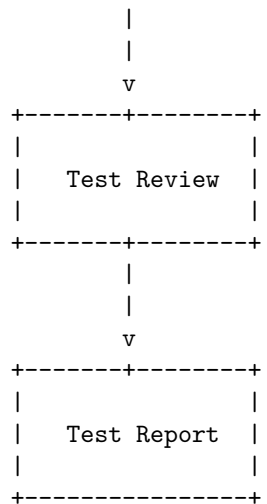


The process starts with **Review Planning**, where the objectives, scope, and participants of the review are determined. Next, the **Review Meeting** takes place, where the participants discuss the work product and identify any issues or defects. After the meeting, any necessary **Rework & Follow-up** is performed to address the identified issues. The results of the review are then documented in the **Review Reporting** phase. Finally, the review is considered complete in the **Review Completion** phase.

### Walk Through (Walkthrough) Static testing strategy

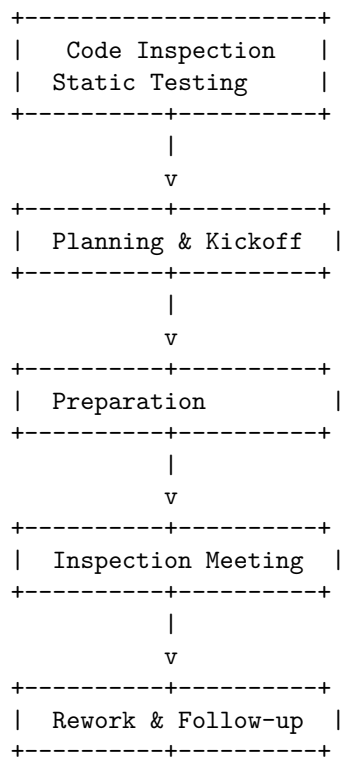
Here is an ASCII diagram that represents the Walk Through Static testing strategy:





### Code Inspection (Code Inspection) Static testing strategy

Here is an ASCII diagram that represents the Code Inspection (Code Inspection) Static testing strategy:



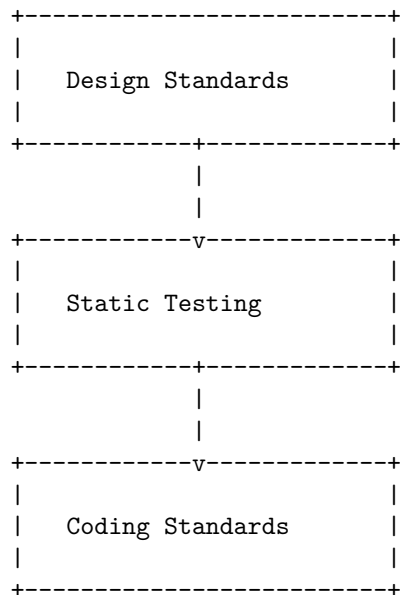
```

      |
      v
+-----+
|  Inspection Complete  |
+-----+

```

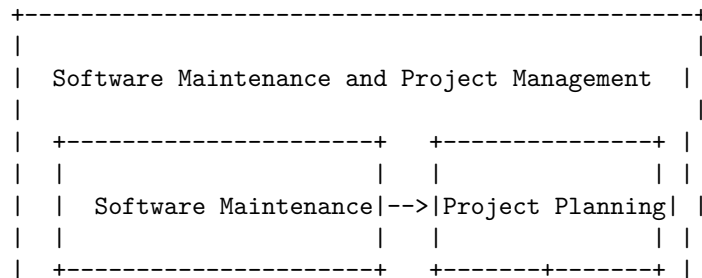
### Compliance with Design and Coding Standards (Coding Standards) Static testing strategy

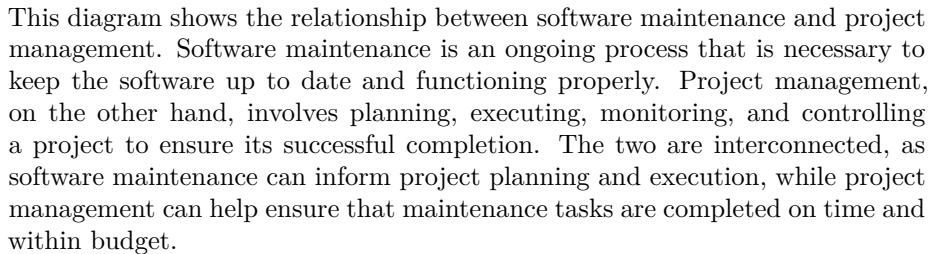
Here is an ASCII diagram that represents the static testing strategy for compliance with design and coding standards:



### Unit 5 - Software Maintenance and Software Project Management

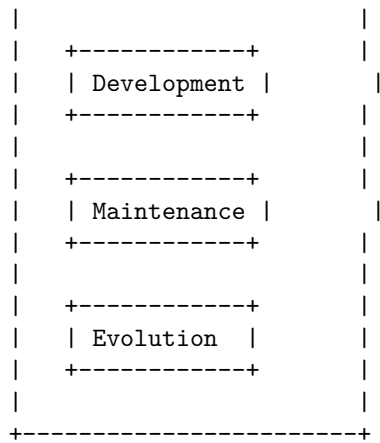
Here is an ASCII diagram for Unit 5 - Software Maintenance and Software Project Management:



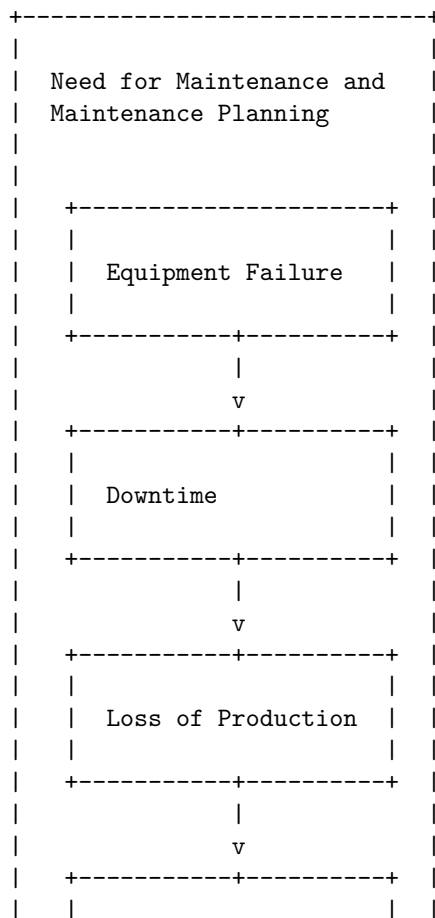


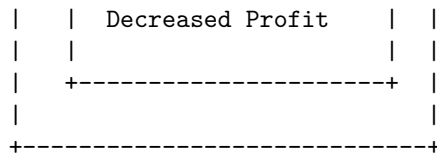
Here is an ASCII diagram that represents the concept of software as an evolutionary entity:





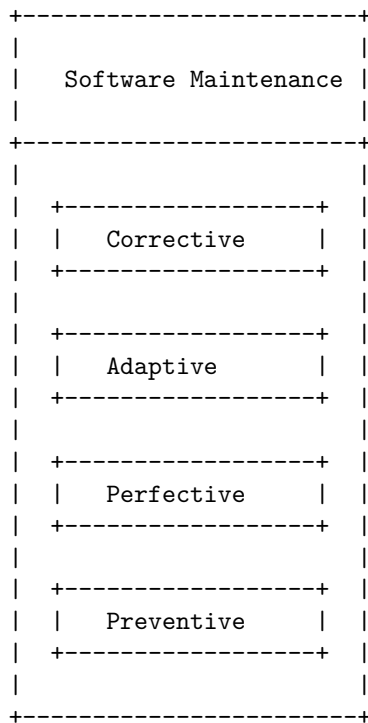
## Need for Maintenance and Maintenance Planning





### Categories of Maintenance of Software

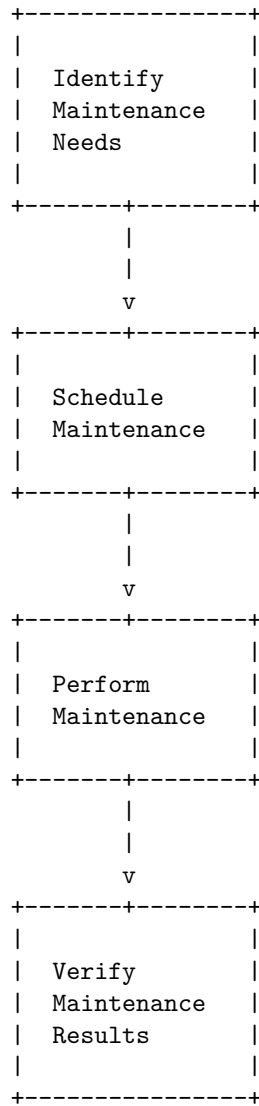
There are four main categories of software maintenance: corrective, adaptive, perfective, and preventive. Here is an ASCII diagram that illustrates these categories:



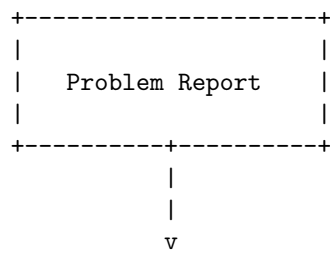
Corrective maintenance involves fixing errors and defects in the software. Adaptive maintenance involves making changes to the software to accommodate changes in the environment, such as new hardware or operating systems. Perfective maintenance involves making changes to the software to improve its performance or usability. Preventive maintenance involves making changes to the software to prevent future problems or defects.

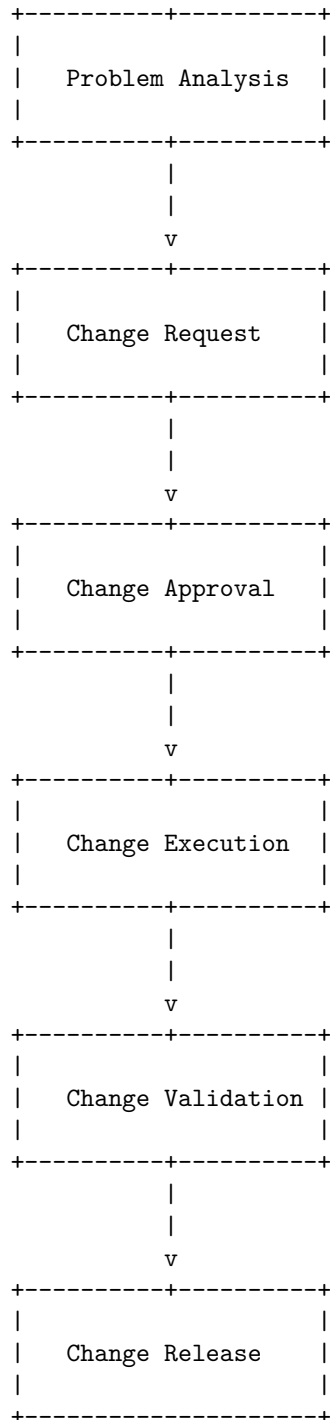
### Preventive Maintenance (PM) of Software

Here is an ASCII diagram that illustrates the process of Preventive Maintenance (PM) of Software:

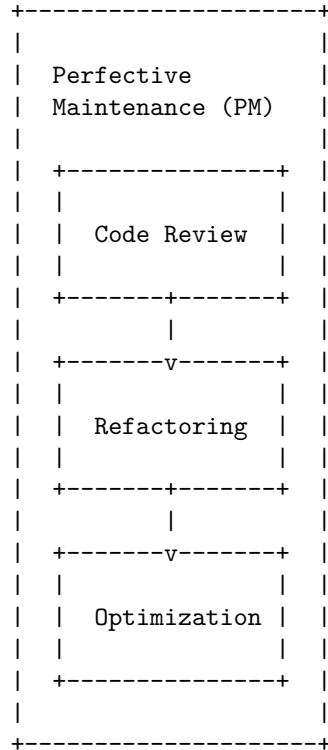


#### Corrective Maintenance (CM) of Software



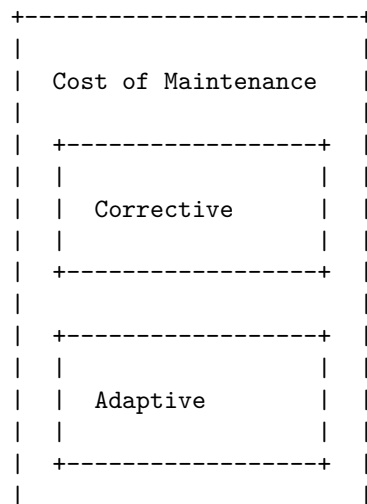


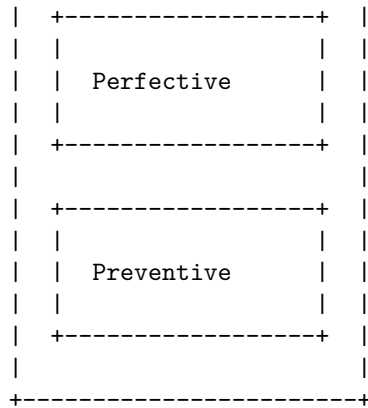
## Perfective Maintenance (PM) of Software



## Cost of Maintenance of Software

Here is an ASCII diagram that shows the cost of maintenance of software:

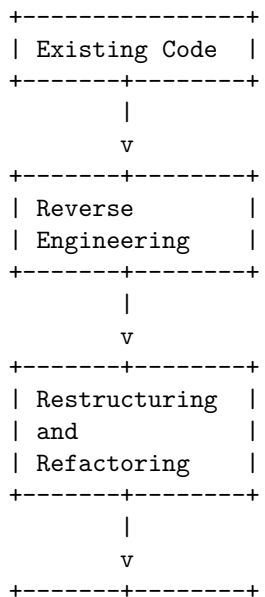


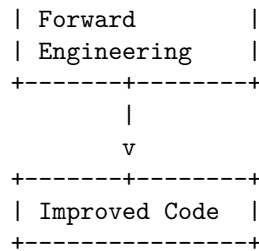


The cost of maintenance of software can be divided into four categories: Corrective, Adaptive, Perfective, and Preventive. Corrective maintenance involves fixing errors and bugs in the software. Adaptive maintenance involves making changes to the software to adapt to changes in the environment. Perfective maintenance involves improving the performance or maintainability of the software. Preventive maintenance involves taking steps to prevent future problems with the software.

### Software Re- Engineering (SR) of Software

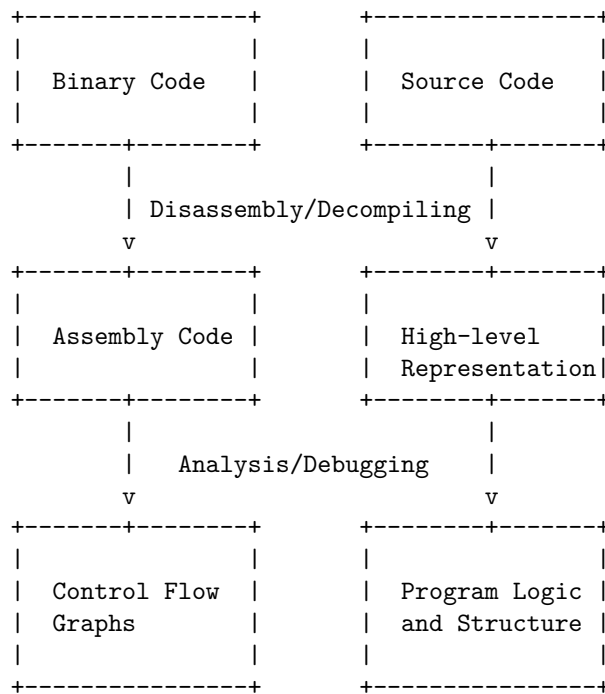
Software re-engineering is the process of improving the design, structure, and implementation of existing software systems while preserving their functionality. Here is an ASCII diagram that illustrates the process of software re-engineering:





### Reverse Engineering (RE) of Software

Here is an ASCII diagram that illustrates the process of reverse engineering (RE) of software:



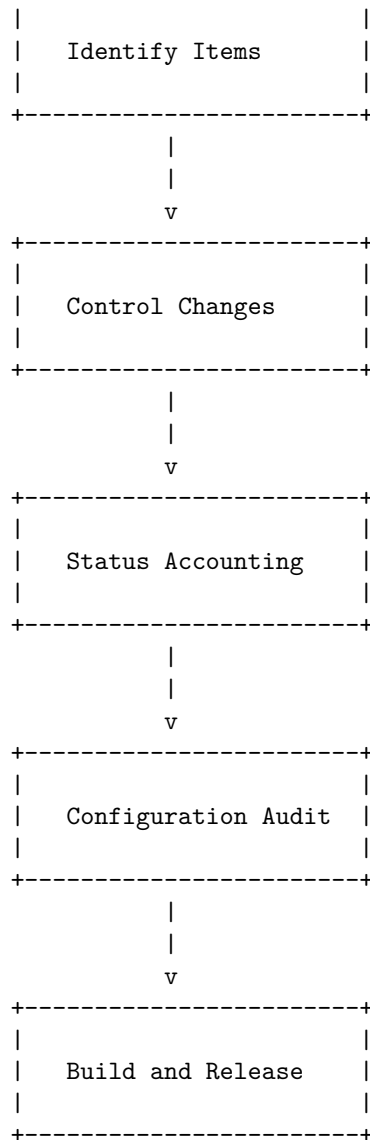
In the process of reverse engineering, the binary code of the software is disassembled or decompiled to produce assembly code or a high-level representation of the code. This code is then analyzed and debugged to produce control flow graphs and an understanding of the program logic and structure. This information can be used to modify or improve the software, or to develop new software that is compatible with the original.

### Software Configuration Management Activities

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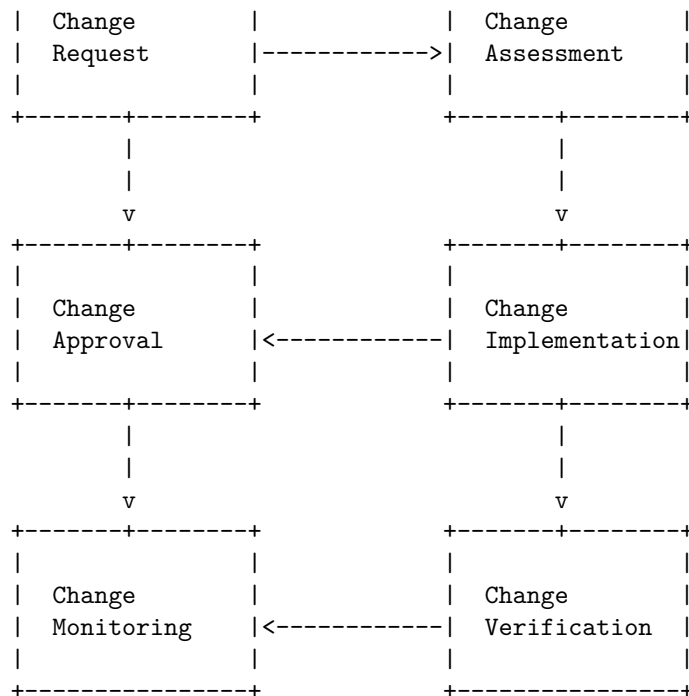
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### Change Control Process in software project management

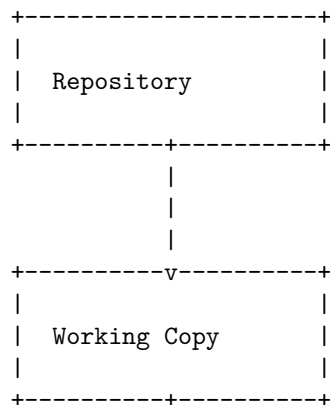
The Change Control Process in software project management is a formal process used to ensure that changes to a product or system are introduced in a controlled and coordinated manner. Here is an ASCII diagram that illustrates the steps involved in the Change Control Process:

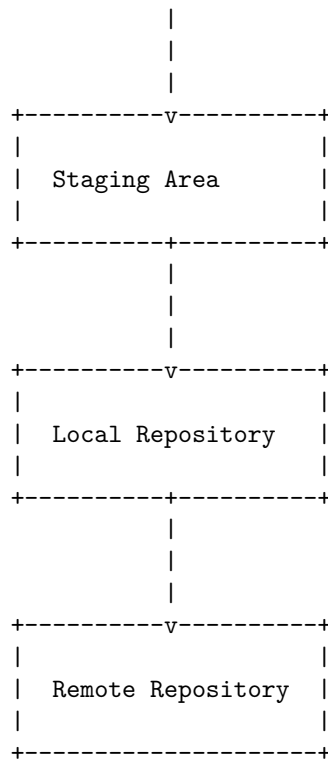




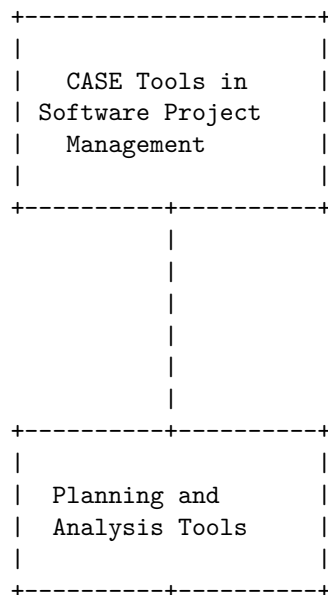
The process begins with a Change Request, which is a proposal for an alteration to the product or system. This request is then assessed to determine its impact and feasibility. If the change is approved, it is then implemented. After implementation, the change is verified to ensure that it has been implemented correctly and is functioning as intended. Finally, the change is monitored to ensure that it continues to function correctly and meet the needs of the organization.

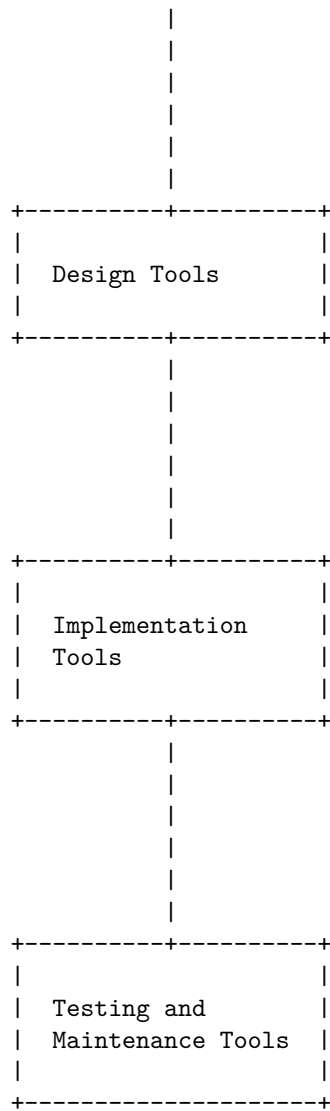
### Software Version Control in software project management





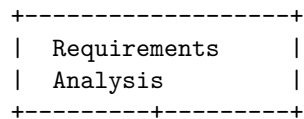
### An Overview of CASE Tools in software project management



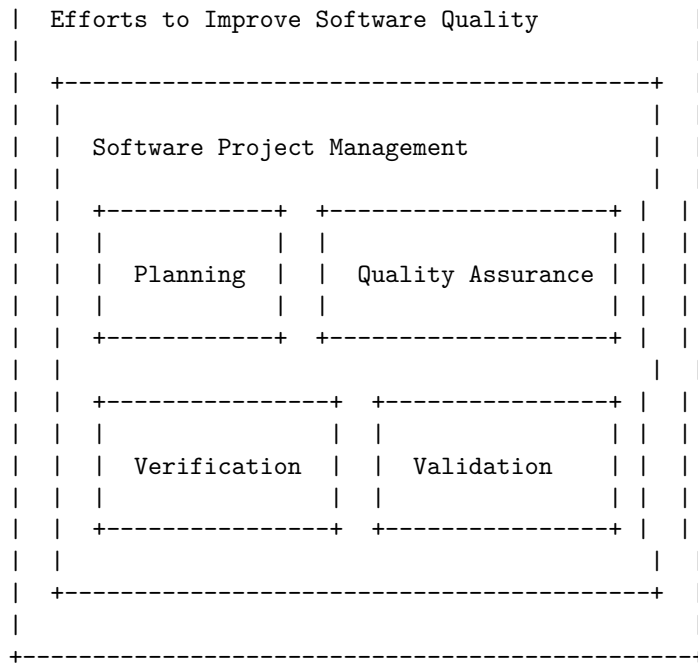


### Estimation of Various Parameters such as Cost and Time in software project management

Here is an ASCII diagram that represents the process of estimating various parameters such as cost and time in software project management:







#### Schedule/Duration of Maintenance in software project management

| Planning Phase                                                                                                                                     | Execution Phase                                                                                                         | Monitoring Phase                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Define scope</li> <li>- Identify risks</li> <li>- Develop schedule</li> <li>- Assign resources</li> </ul> | <ul style="list-style-type: none"> <li>- Implement changes</li> <li>- Test changes</li> <li>- Deploy changes</li> </ul> | <ul style="list-style-type: none"> <li>- Monitor progress</li> <li>- Report status</li> <li>- Identify issues</li> <li>- Implement corrective actions</li> </ul> |

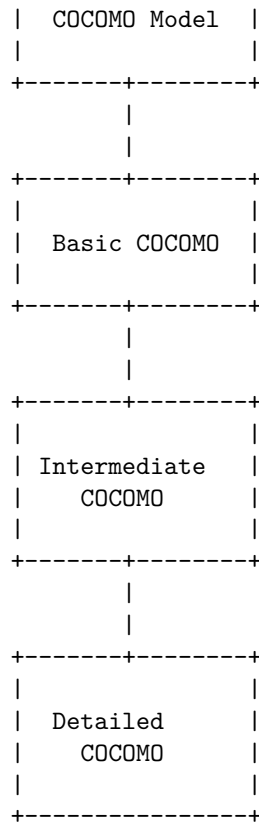
#### Constructive Cost Models (COCOMO) in software project management

Here is an ASCII diagram for the Constructive Cost Models (COCOMO) in software project management:

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|               |

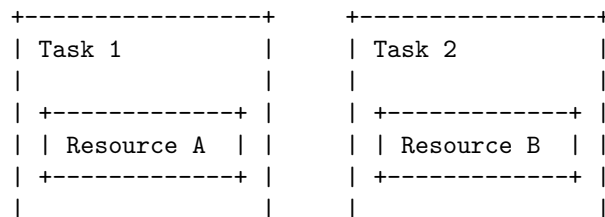
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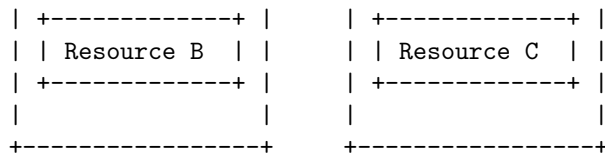


The COCOMO model is a hierarchy of three increasingly detailed and accurate forms. The first level, Basic COCOMO, is good for quick, early, rough order of magnitude estimates of software costs. The second level, Intermediate COCOMO, takes into account a number of cost drivers for software development. The third level, Detailed COCOMO, breaks the software development process down into a number of activities and uses different cost drivers for each activity.

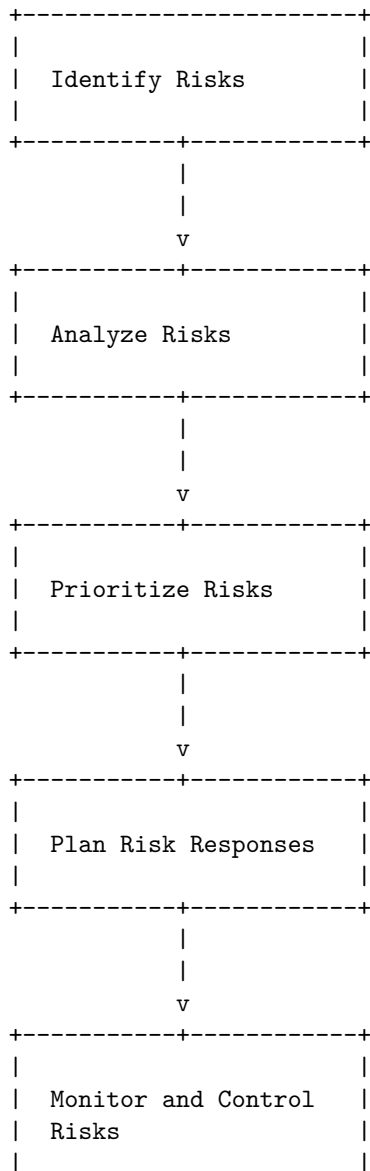
### Resource Allocation Models (RAIM) in software project management

Resource Allocation Models (RAIM) are used in software project management to allocate resources to tasks in a project. Here is an ASCII diagram that represents a simple RAIM:





## Software Risk Analysis and Management in software project management



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## Software Project Management

Software project management is the process of planning, organizing, and managing resources to successfully complete software development projects. It involves the following key activities:

1. **Project Planning:** This involves defining the scope of the project, identifying the tasks that need to be completed, estimating the time and resources required, and creating a project schedule.
2. **Resource Management:** This involves identifying the resources required for the project, including personnel, hardware, and software, and allocating them appropriately.
3. **Risk Management:** This involves identifying potential risks that could impact the project, assessing their likelihood and potential impact, and developing strategies to mitigate or avoid them.
4. **Quality Management:** This involves ensuring that the software meets the specified requirements and standards, and that it is delivered on time and within budget.
5. **Communication Management:** This involves ensuring that all stakeholders are kept informed of the project's progress, and that any issues or concerns are addressed in a timely manner.

Effective software project management is essential for the successful completion of software development projects. It helps to ensure that the project is completed on time, within budget, and to the required quality standards. It also helps to minimize risks and maximize the chances of success.